

6.3 Transition Element Complexes: Isomers, Reactions & Stability (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	6. Inorganic Chemistry (A Level Only)
Topic	6.3 Transition Element Complexes: Isomers, Reactions & Stability (A Level Only)
Difficulty	Easy

Time allowed: 30

Score: /20

Percentage: /100

Question 1a

This question is about ligand exchange in copper(II) complexes.

Complete Table 1.1 to show whether the coordination number and shape of the complex changes in the given situations.

Table 1.1

	Ligand exchange of similarly sized ligands	Ligand exchange of differently sized ligands
Coordination number		
Shape		

[2 marks]**Question 1b**

When copper(II) sulfate dissolves in water, the conventional equation for this can be represented as



State the formula of the hexaaquacopper(II) complex represented by $\text{Cu}^{2+}(\text{aq})$.

[1 mark]

Question 1c

Complete the three-dimensional diagram in Fig. 1.1 to show the hexaaquacopper(II) ion complex.

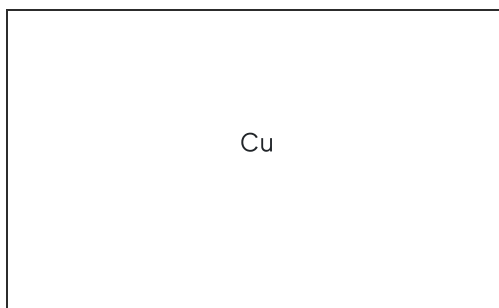


Fig. 1.1

[2 marks]

Question 1d

Hexaaquacopper(II) ions react with concentrated hydrochloric acid to form complex **A**.

Complete Table 1.2 with the colour and geometry of the hexaaquacopper(II) complex and the colour, geometry and formula of complex **A**.

Table 1.2

	Hexaaquacopper(II) complex	A
colour of complex		
geometry of complex		
formula of complex		

[2 marks]

Question 1e

In the presence of concentrated ammonia, hexaaquacopper(II) initially reacts to form $\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4$.

i)

Give the colour and state of $\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4$.

colour

state

[1]

ii)

In the presence of excess concentrated ammonia, the $\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4$ further reacts to form $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$.

Give the colour and state of $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$.

colour

state

[1]

iii)

The $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ formed is a mixture of two geometrically isomeric octahedral complexes.

Fig. 1.2 shows the three-dimensional diagram of the trans isomer. Complete Fig. 1.2 to show the three-dimensional diagram of the cis isomer.

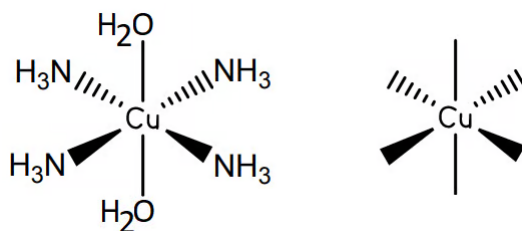


Fig. 1.2

[1]

[3 marks]

Question 2a

The feasibility of redox reactions can be determined using standard electrode potential, E^θ , values.

The following electrochemical cell is set up.

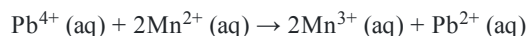


Table 2.1

Electrode reaction	E^θ / V
$\text{Mn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mn}$	- 1.18
$\text{Mn}^{3+} + \text{e}^- \rightleftharpoons \text{Mn}^{2+}$	+ 1.49
$\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb}$	- 0.13
$\text{Pb}^{4+} + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+}$	+ 1.69

Use Table 2.1 to determine the half-equations for reduction and oxidation. Your half-equations should be written in the conventional reduction format.

Reduction half-equation

Oxidation half-equation

[2 marks]

Question 2b

Use Table 2.1 and your answer to part (a) to calculate the standard cell potential, E^θ_{cell} , for the reaction.

$E^\theta_{\text{cell}} = \dots\dots\dots \text{V}$

[1 mark]

Question 2c

Use your answer to part (b) to suggest whether the reaction is feasible. Explain your answer.

[1 mark]

Question 3a

Potassium manganate(VII) can be used to titrate a sample containing an unknown percentage of iron(II).

- i)
Complete the half-equation for the reduction of manganate ions, MnO_4^- , to manganese ions, Mn^{2+} .



[1]

- ii)
Explain whether this reaction is oxidation or reduction.

[1]

[2 marks]

Question 3b

- i)
Write the half-equation for the oxidation of Fe^{2+} to Fe^{3+} . Your answer should include state symbols.

[1]

- ii)
Explain whether this reaction is oxidation or reduction.

[1]

[2 marks]

Question 3c

Using your answers to parts **(a)** and **(b)**, write the full equation for the reaction between Fe^{2+} ions and MnO_4^- ions.

[2 marks]

Model Answers: Easy

1a

a) The completed table is:

	Ligand exchange of similarly sized ligands	Ligand exchange of differently sized ligands	
Coordination number	No change	No change	; [1 mark]
Shape	Change	Change	; [1 mark]

[Total: 2 marks]

- As the ligands become larger, fewer of them can fit around the central metal ion
 - This reduces the number of dative covalent / coordinate bonds to the central metal ion or the coordination number
- Changing the number of ligands bonded to the central metal ion also results in a change in shape

1b

b) The formula of the hexaaquacopper(II) complex represented by $\text{Cu}^{2+}(\text{aq})$ is:

- $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$; [1 mark]

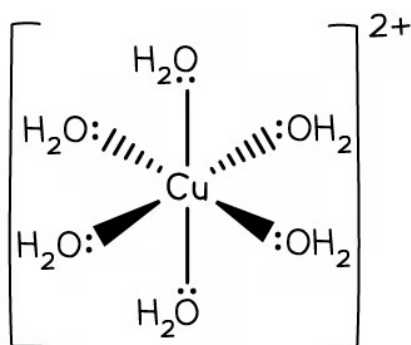
[Total: 1 mark]

- The only difference between the formula of the $\text{Cu}^{2+}(\text{aq})$ ion in the question and the formula of the hexaaquacopper(II) ion is the inclusion of 6 water ligands to form the complex
 - The hexaaquacopper(II) name tells you that there are six (hexa) water molecules (aqua) complexed to a copper(II) / Cu^{2+} ion
- Therefore, the answer is $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$
- Remember:** Charged complexes should have square brackets around them

- **Careful:** Even if the question doesn't say, you should include state symbols for transition metal complexes as examiners usually take a mark if they are missing or wrong

1c

c) The completed three-dimensional diagram to show the hexaaquacopper(II) ion complex is:



- Central copper ion surrounded by six water molecules
AND
Two of each bond type; [1 mark]
- Square brackets around the complex
AND
2+ charge outside the brackets; [1 mark]

[Total: 2 marks]

- From the previous question, we know that there are six water ligands / molecules complexed to a Cu^{2+} ion
- Place the Cu^{2+} ion in the centre and surround it with the six water molecules / ligands
- Since you are asked for the three-dimensional diagram:
 - Add two straight bonds between the Cu and two water ligands / molecules
 - Add two dotted bonds between the Cu and two water ligand / molecules
 - Add two wedge bonds between the Cu and two water ligand / molecules
- Add square brackets and the charge to your complex

1d

d) The completed table is:

	Hexaaquacopper(II) complex	A
colour of complex	blue / pale blue	yellow
geometry of complex	octahedral	tetrahedral
formula of complex		$[\text{CuCl}_4]^{2-}$

- Any three cells correct; [1 mark]
- All five cells correct; [2 marks]

[Total: 2 marks]

- You are expected to know the ligand exchange reactions for complexes of copper(II) ions and cobalt(II) ions with water and ammonia molecules and hydroxide and chloride ions
 - This knowledge should include the colours, geometry, formulae and how to write ligand exchange equations

1e

e)

i) The colour and state of $\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4$ are:

- Colour = pale blue

AND

State = solid / (s) / precipitate; [1 mark]

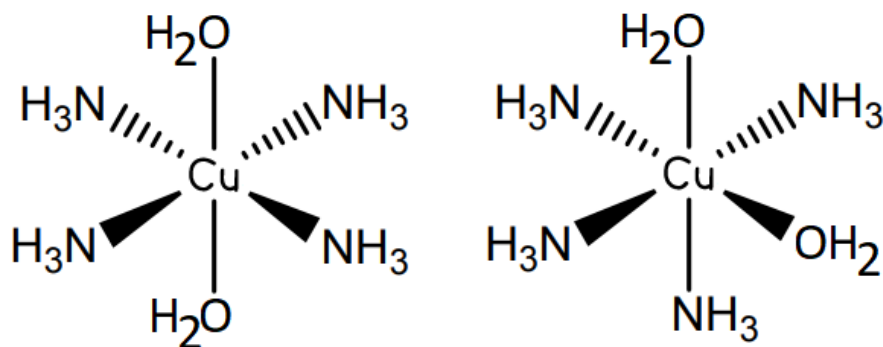
ii) The colour and state of $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$.

- Colour = deep / dark / royal blue

AND

State = aqueous / (aq); [1 mark]

iii) The completed three-dimensional diagram to show the cis isomer of $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ is:



- Correct structure of the cis-isomer (right-hand side); [1 mark]

[Total: 3 marks]

- You are expected to know the ligand exchange reactions of copper(II) ion complexes with ammonia
 - This knowledge should include the colours, geometry, formulae and how to write ligand exchange equations
- The trans-isomer structure that you are given shows the two water molecules opposite each other
 - The cis-isomer will have the two water molecules adjacent to each other

2a

a) The half-equations for reduction and oxidation are:

- Reduction half-equation = $\text{Pb}^{4+} + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+}$; [1 mark]
- Oxidation half-equation = $\text{Mn}^{3+} + \text{e}^- \rightleftharpoons \text{Mn}^{2+}$; [1 mark]

[Total: 2 marks]

- From the equation given in the question, you should be able to identify that the required half-equations involve:
 - $\text{Pb}^{4+} + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+}$
 - This is the reduction half-equation because it has the most positive standard electrode potential
 - $\text{Mn}^{3+} + \text{e}^- \rightleftharpoons \text{Mn}^{2+}$
 - This is the oxidation half-equation because it has the least positive standard electrode potential

2b

b) The standard cell potential, E^θ_{cell} , for the reaction is:

- + 0.20 (V); [1 mark]

[Total: 1 mark]

- **Remember:** $E^\theta_{\text{cell}} = E^\theta_{\text{reduction}} - E^\theta_{\text{oxidation}}$
 - $E^\theta_{\text{cell}} = (+1.69) - (+1.49) = +0.20 \text{ V}$

2c

c) The reaction is:

- Feasible

AND

Because the standard cell potential / E^θ_{cell} is positive; [1 mark]

[Total: 1 mark]

- **Remember:** A positive value of E^θ_{cell} means that a reaction is feasible, while a negative value means that the reaction is not feasible

3a

a)

i) The complete half-equation for the reduction of manganate ions, MnO_4^- , to manganese ions, Mn^{2+} is:

- $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{aq})$; [1 mark]

ii) The reaction is:

- Reduction

AND

Because there is a gain of electrons; [1 mark]

[Total: 2 marks]

- To balance the manganate half-equation:
 - $\text{MnO}_4^- (\text{aq}) + \dots\dots\text{H}^+ (\text{aq}) + \dots\dots\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + \dots\dots\text{H}_2\text{O} (\text{aq})$
 - The four oxygens in MnO_4^- will make $4\text{H}_2\text{O}$
 - $\text{MnO}_4^- (\text{aq}) + \dots\dots\text{H}^+ (\text{aq}) + \dots\dots\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{aq})$
 - The four water molecules will require 8H^+
 - $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + \dots\dots\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{aq})$
 - Once the atoms are balanced, the charges need to be balanced
 - The left-hand side has an overall charge of +7, not including the electrons that will be used to balance the charge
 - The right-hand side has an overall charge of +2
 - Therefore, 5e^- are required
 - $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{aq})$
- **Remember:** OIL-RIG: Oxidation is loss, reduction is gain of electrons

3b

b)

i) The half-equation for the oxidation of Fe^{2+} to Fe^{3+} is:

- $\text{Fe}^{2+} (\text{aq}) \rightarrow \text{Fe}^{3+} (\text{aq}) + \text{e}^-$; [1 mark]

ii) This reaction is :

- Oxidation

AND

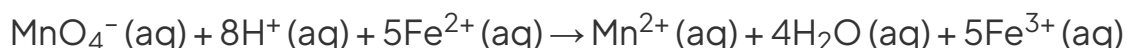
Because electrons are lost; [1 mark]

[Total: 2 marks]

- Compare to part (a), the Fe^{2+} half-equation is simpler because it is losing one electron to become Fe^{3+}
 - $\text{Fe}^{2+} (\text{aq}) \rightarrow \text{Fe}^{3+} (\text{aq}) + \text{e}^-$
 - **Careful:** The question instructs you to include state symbols
 - You should be including state symbols as a matter of good practice and avoid losing a potential mark in any question
- **Remember:** OIL-RIG: Oxidation is loss, reduction is gain

3c

c) The full equation for the reaction between Fe^{2+} ions and MnO_4^- ions is:



- Correct reactants and products; [1 mark]
- Correct balancing; [1 mark]

[Total: 2 marks]

- The MnO_4^- half-equation is:
 - $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{aq})$
- The Fe^{2+} half-equation is:
 - $\text{Fe}^{2+} (\text{aq}) \rightarrow \text{Fe}^{3+} (\text{aq}) + \text{e}^-$
- You need to multiply the Fe^{2+} equation by 5 so that the number of electrons in both half-equations have the same number of electrons
 - $5\text{Fe}^{2+} (\text{aq}) \rightarrow 5\text{Fe}^{3+} (\text{aq}) + 5\text{e}^-$
- Combining the two half-equations gives:
 - $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{Fe}^{2+} (\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{aq}) + 5\text{Fe}^{3+} (\text{aq}) + 5\text{e}^-$
- The electrons on both sides cancel out to give the final overall answer:
 - $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{Fe}^{2+} (\text{aq}) \rightarrow \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{aq}) + 5\text{Fe}^{3+} (\text{aq})$



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